Information Security

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Motivation


فَبِاَيِّ اٰلَآءِ رَبِّكُمَا تُكَذِّبٰنِ
تو (اے جن و انسان!) تم دونوں اپنے رب کی کون کون سی نعمتوں کو جھٹلاؤ گے؟
SURAH REHMAN 55:13

Cryptology

Cryptology

"Study of techniques for ensuring the secrecy and authenticity of information"

The main branches of cryptology are:-

- Cryptography
- Cryptanalysis

Cryptography

- Cryptography comes from Greek words meaning “hidden writing”
- Study the design of such techniques
- preventing private data from being stolen
- Cryptography is the practice and study of techniques for secure communication in the **presence of third parties**, often referred to as opponents.

Cryptography

Generically classified along three independent dimensions

- Type of **operations** used for transforming plaintext to ciphertext
- Number of **Keys** used
- **Way** in which plaintext is processed

Cryptography

Core **goals** are CIA & Non-repudiation

The three major types of algorithms used for encryption are:

- ✓ **Symmetric**, also called private or secret key
- ✓ **Asymmetric**, also called public key
- ✓ Hashing

Cryptography

Applications of Cryptography

- Secure communication (e.g., messaging apps, VPNs).
- Blockchain and cryptocurrencies (e.g., Bitcoin, Ethereum).
- Password storage and authentication.
- Secure voting systems and e-commerce.

Cryptanalysis

- Cryptanalysis is the study of analyzing and **breaking cryptographic systems** to **uncover weaknesses or vulnerabilities.**
- It involves understanding how **cryptographic algorithms work and exploiting flaws in their design, implementation, or usage.**

Cryptosystem

The combination of algorithm, key, and key management functions used to perform cryptographic operations

Cryptographic Tools

- cryptographic algorithms are important element in security services
- review various types of elements
  - symmetric encryption
  - public-key (asymmetric) encryption
  - digital signatures and key management
  - secure hash functions
- example is use to encrypt stored data

Cryptography

Symmetric (single key, Secret Key, Conventional Key)

Both sender and receiver use the same key

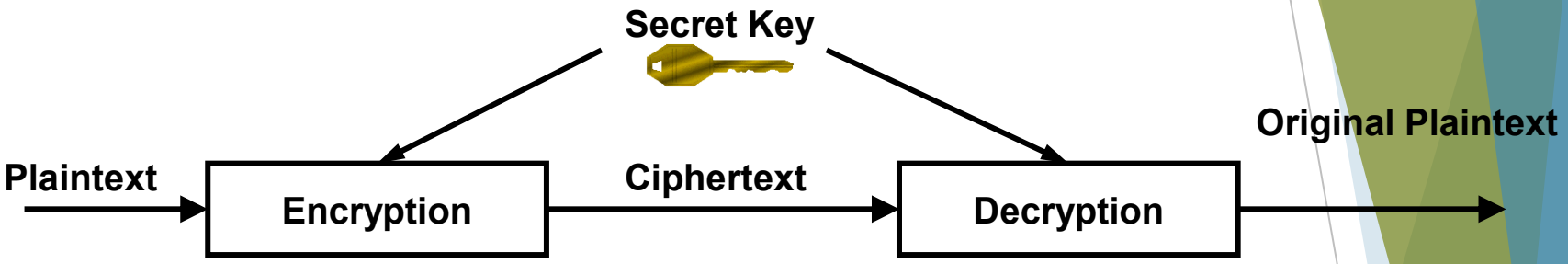
Asymmetric (two-keys, or public-key encryption)

Sender and receiver use a different key

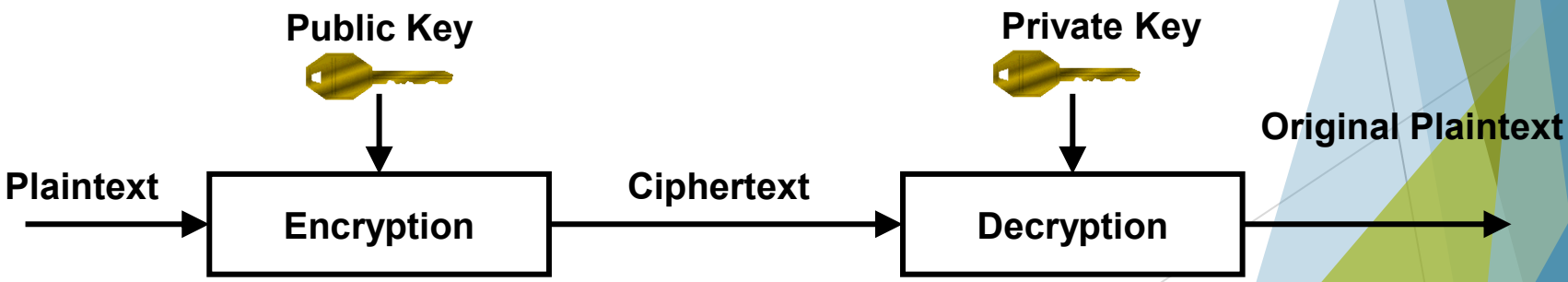
Comparison of SK and PK Cryptography

DISTINCT FEATURES	SECRET KEY	PUBLIC KEY
NUMBER OF KEYS	Single key	Pair of keys
TYPES OF KEYS	Key is secret	One key is private, and one key is public
LENGTH OF KEYS	40-200 bits	512-2048 bits
RELATIVE SPEEDS	Faster	Slower

Comparison of SK and PK Cryptography



Symmetric (Single Key) Cryptography



Asymmetric (Two Key) Cryptography

Symmetric Encryption

- The universal technique for providing confidentiality for transmitted data is symmetric encryption.
- Symmetric encryption is a type of encryption where only one key (a secret key) is used to both encrypt and decrypt electronic information. By using symmetric encryption algorithms, data is converted to a form that cannot be understood by anyone who does not possess the secret key to decrypt it
- A symmetric encryption scheme has five ingredients

Symmetric Cipher Model (Another view)

Symmetric Cipher schemes have Five ingredients:-

Plaintext original, intelligent message Or data fed into algorithm as input

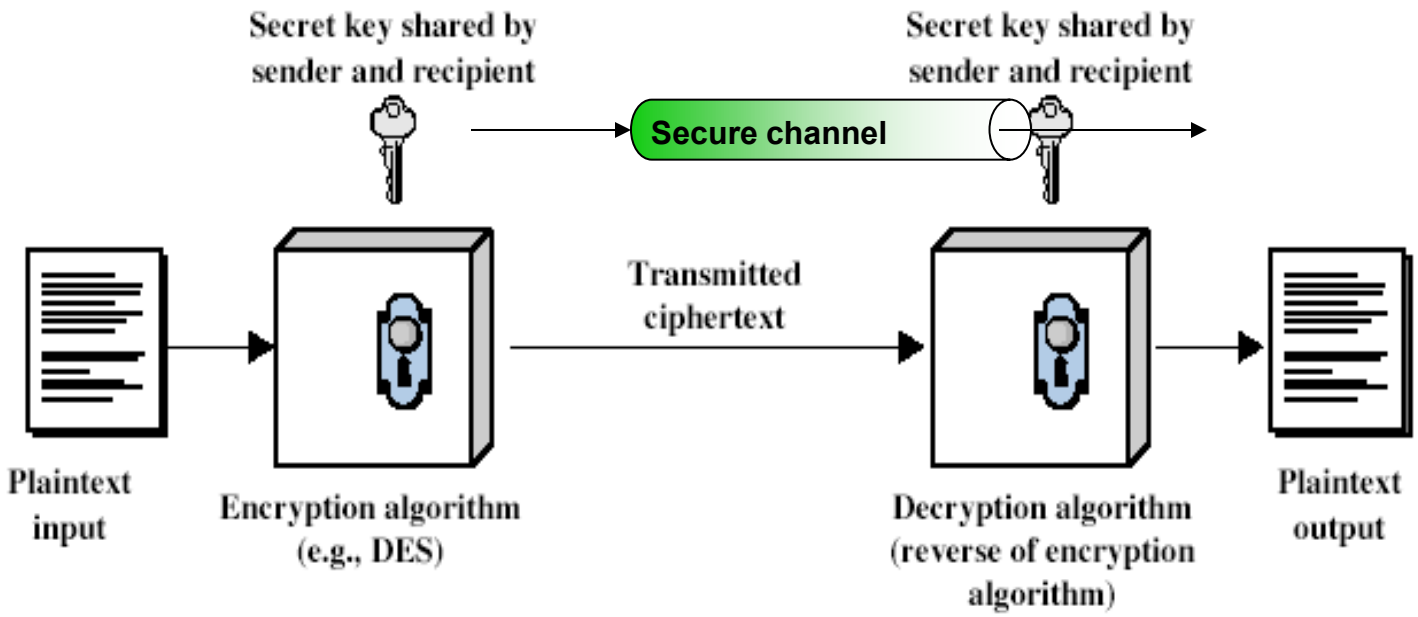
Encryption Algorithm (cipher) algorithm for transforming plaintext to ciphertext

Ciphertext coded message or scrambled message as an output

Secret Key Info used in cipher known only to sender/receiver

Decryption Algorithm (cipher) algorithm for transforming ciphertext to plaintext

Symmetric Cipher Model



Cryptanalysis

Objective to recover key not just message

General approaches are:-

- Cryptanalytic attack
- Brute-force attack

Brute Force Attack

When encryption and decryption algorithms are publicly available, as they generally are, a brute-force attack means trying every possible key on a piece of ciphertext until an intelligible translation into plaintext is obtained.

Cryptanalytic Attacks

Cryptanalytic Attacks depends on

- known algorithm
- Knowledge of the general characteristics of the plaintext even sample plaintext & ciphertext
- Attacker will try to deduce the key is being used.

Brute Force Search

- Always possible to simply try every key
- Most basic attack, proportional to key size

Key Size (bits)	Number of Alternative Keys	Time required at 1 decryption/μs	Time required at 10^6 decryptions/μs
32	$2^{32} = 4.3 \times 10^9$	2^{31} μs = 35.8 minutes	2.15 milliseconds
56	$2^{56} = 7.2 \times 10^{16}$	2^{55} μs = 1142 years	10.01 hours
128	$2^{128} = 3.4 \times 10^{38}$	2^{127} μs = 5.4×10^{24} years	5.4×10^{18} years
168	$2^{168} = 3.7 \times 10^{50}$	2^{167} μs = 5.9×10^{36} years	5.9×10^{30} years
26 characters (permutation)	$26! = 4 \times 10^{26}$	2×10^{26} = 6.4×10^{12} years	6.4×10^6 years

Cryptography

Block cipher

"Encrypts/decrypts a block at a time"

- Size of Plaintext=Size of ciphertext
- Typically block size is 64-bit or 128
- Examples of Block Cipher are DES and Feistel Cipher Structure

Stream cipher

"Encrypts/decrypts one element at a time (Continuously inputs)"

- It encrypts a digital data stream one bit or one byte at a time (bit by bit or byte by byte)
- Examples of Stream Cipher are the autokeyed Vigener Cipher and the Vernam Cipher

Unconditional & Computational Security

A cipher is unconditionally secure if it is secure no matter how much resources (time, space) the attacker has.

A cipher is computationally secure if the best algorithm for breaking it will require so much resources (e.g., 1000 years) that practically the cryptosystem is secure.

Note:-All the ciphers we have examined are not unconditionally secure.

Steganography

- Greek word means ”Concealed Writing ”
- An alternative to encryption
- Hides existence of message
  - using invisible ink
  - hiding in graphic image or sound file
  - Microdots
  - Concealing messages within the lowest bits of a digital image
  - Tattoo a message
- **Drawbacks**
  - high overhead to hide relatively few info bits

Reference

[1] William Stallings,(2011).Cryptography and Network Security! New Jersey : Prentice Hall.

[2] Douglas Stinson ,(1995).Cryptography -Theory and Practice Boca Raton.

[3] Computer Security: Principles and Practice, 3rd edition by William Stallings

Jazak Allah Khair